

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: data=pd.read_csv(r'C:\Users\hp\Downloads\myexcel - myexcel.csv.csv')
data
```

Out[2]:

	Name	Team	Number	Position	Age	Height	Weight	College	Salary
0	Avery Bradley	Boston Celtics	0	PG	25	06-Feb	180	Texas	7730337.0
1	Jae Crowder	Boston Celtics	99	SF	25	06-Jun	235	Marquette	6796117.0
2	John Holland	Boston Celtics	30	SG	27	06-May	205	Boston University	NaN
3	R.J. Hunter	Boston Celtics	28	SG	22	06-May	185	Georgia State	1148640.0
4	Jonas Jerebko	Boston Celtics	8	PF	29	06-Oct	231	NaN	5000000.0
...	...	...	...	...	...	...	...	...	...
453	Shelvin Mack	Utah Jazz	8	PG	26	06-Mar	203	Butler	2433333.0
454	Raul Neto	Utah Jazz	25	PG	24	06-Jan	179	NaN	900000.0
455	Tibor Pleiss	Utah Jazz	21	C	26	07-Mar	256	NaN	2900000.0
456	Jeff Withey	Utah Jazz	24	C	26	7-0	231	Kansas	947276.0
457	Priyanka	Utah Jazz	34	C	25	07-Mar	231	Kansas	947276.0

458 rows × 9 columns

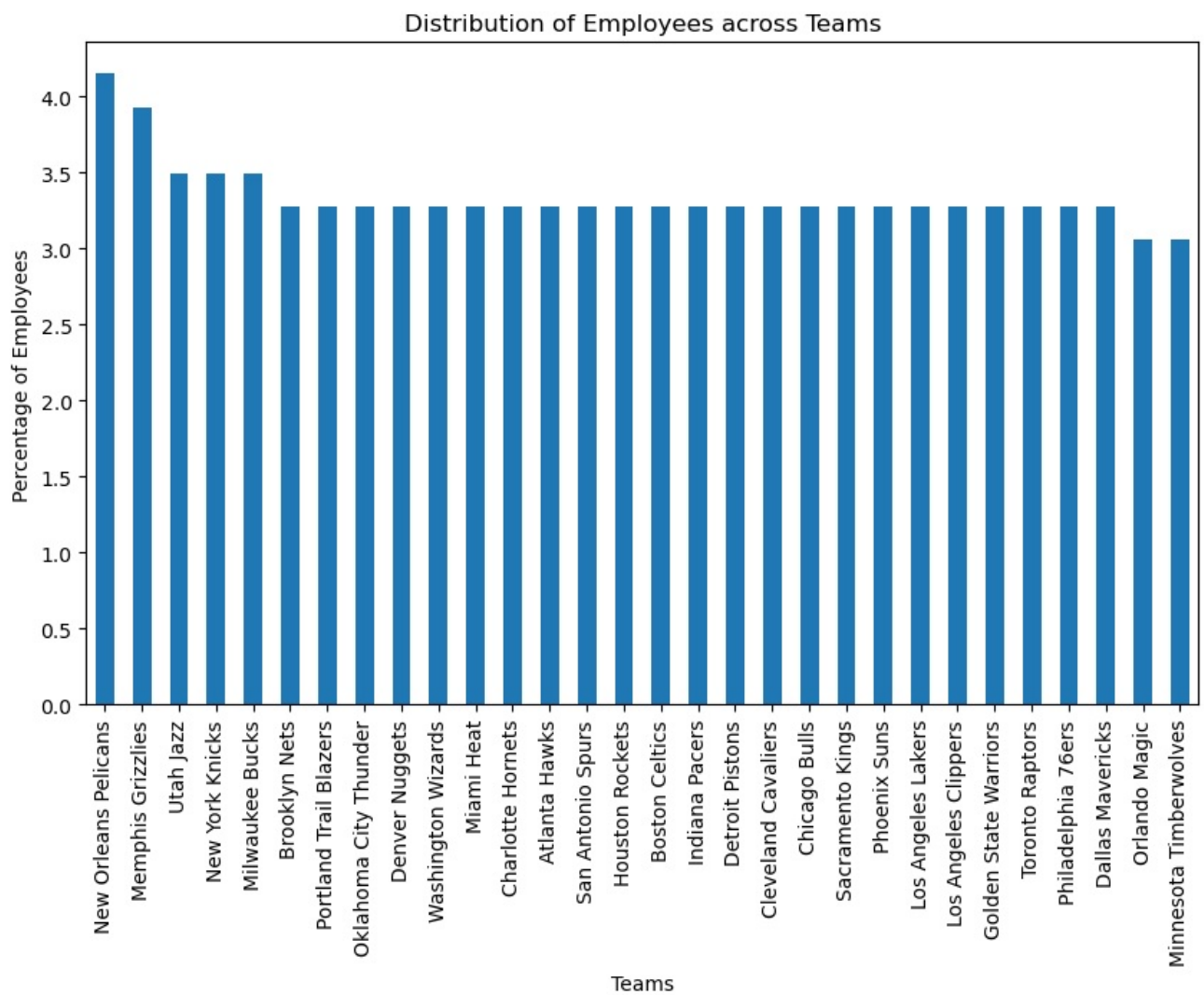
```
In [3]: # Ensure height column exists and replace with random values between 150 and 180
if 'height' in data.columns:
    data['height'] = np.random.randint(150, 181, size=len(df))
```

```
In [4]: team_distribution = data['Team'].value_counts(normalize=True) * 100
team_distribution
```

Out[4]:

Team	
New Orleans Pelicans	4.148472
Memphis Grizzlies	3.930131
Utah Jazz	3.493450
New York Knicks	3.493450
Milwaukee Bucks	3.493450
Brooklyn Nets	3.275109
Portland Trail Blazers	3.275109
Oklahoma City Thunder	3.275109
Denver Nuggets	3.275109
Washington Wizards	3.275109
Miami Heat	3.275109
Charlotte Hornets	3.275109
Atlanta Hawks	3.275109
San Antonio Spurs	3.275109
Houston Rockets	3.275109
Boston Celtics	3.275109
Indiana Pacers	3.275109
Detroit Pistons	3.275109
Cleveland Cavaliers	3.275109
Chicago Bulls	3.275109
Sacramento Kings	3.275109
Phoenix Suns	3.275109
Los Angeles Lakers	3.275109
Los Angeles Clippers	3.275109
Golden State Warriors	3.275109
Toronto Raptors	3.275109
Philadelphia 76ers	3.275109
Dallas Mavericks	3.275109
Orlando Magic	3.056769
Minnesota Timberwolves	3.056769
Name: proportion, dtype: float64	

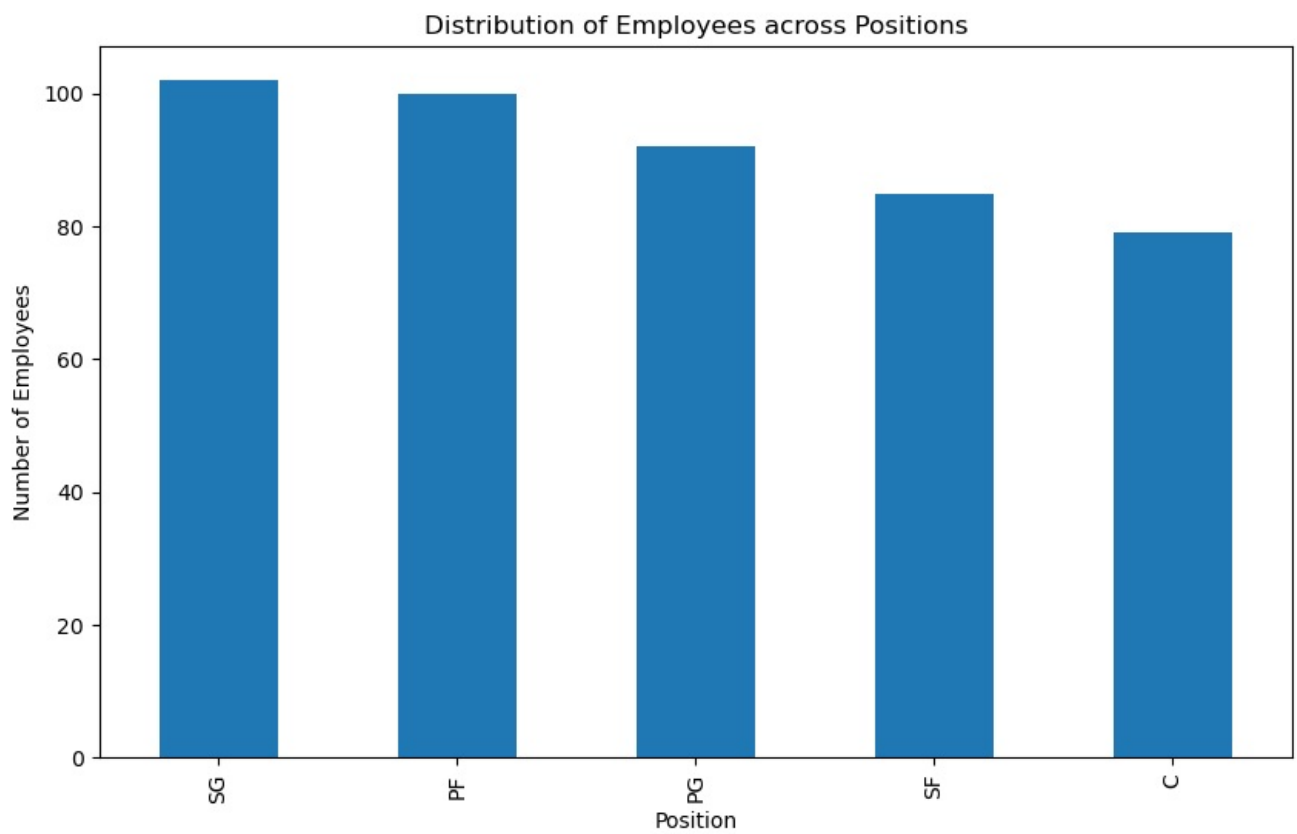
```
In [5]: plt.figure(figsize=(10, 6))
team_distribution.plot(kind='bar')
plt.title('Distribution of Employees across Teams')
plt.ylabel('Percentage of Employees')
plt.xlabel('Teams')
plt.show()
```



```
In [6]: position_distribution = data['Position'].value_counts()
position_distribution
```

```
Out[6]: Position
SG      102
PF      100
PG       92
SF       85
C        79
Name: count, dtype: int64
```

```
In [7]: plt.figure(figsize=(10, 6))
position_distribution.plot(kind='bar')
plt.title('Distribution of Employees across Positions')
plt.ylabel('Number of Employees')
plt.xlabel('Position')
plt.show()
```

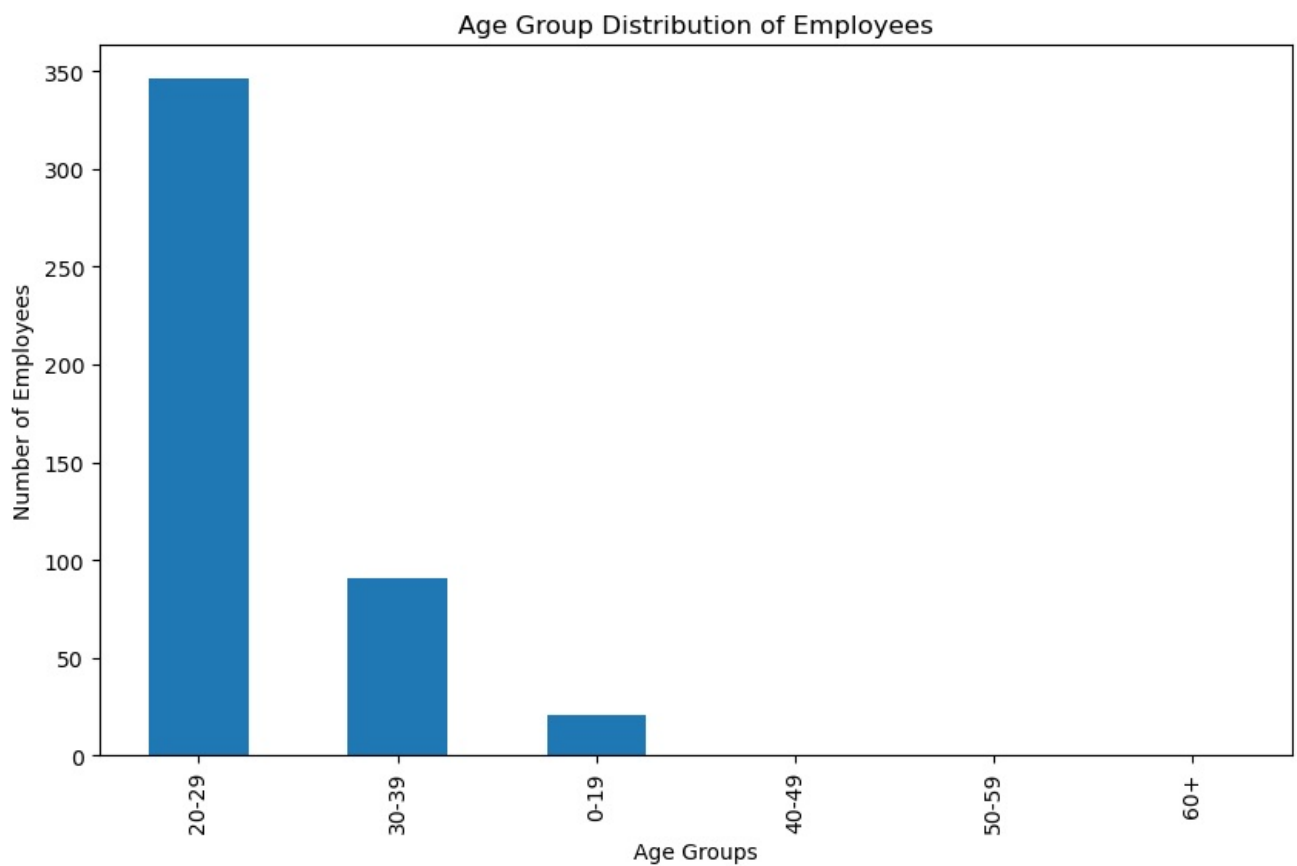


```
In [8]: bins = [0, 20, 30, 40, 50, 60, 100]
labels = ['0-19', '20-29', '30-39', '40-49', '50-59', '60+']
data['Age_group'] = pd.cut(data['Age'], bins=bins, labels=labels)

age_group_distribution = data['Age_group'].value_counts()
age_group_distribution
```

```
Out[8]: Age_group
20-29    346
30-39     91
0-19     21
40-49      0
50-59      0
60+       0
Name: count, dtype: int64
```

```
In [9]: plt.figure(figsize=(10, 6))
age_group_distribution.plot(kind='bar')
plt.title('Age Group Distribution of Employees')
plt.ylabel('Number of Employees')
plt.xlabel('Age Groups')
plt.show()
```



```
In [10]: salary_by_team = data.groupby('Team')['Salary'].sum()
salary_by_position = data.groupby('Position')['Salary'].sum()

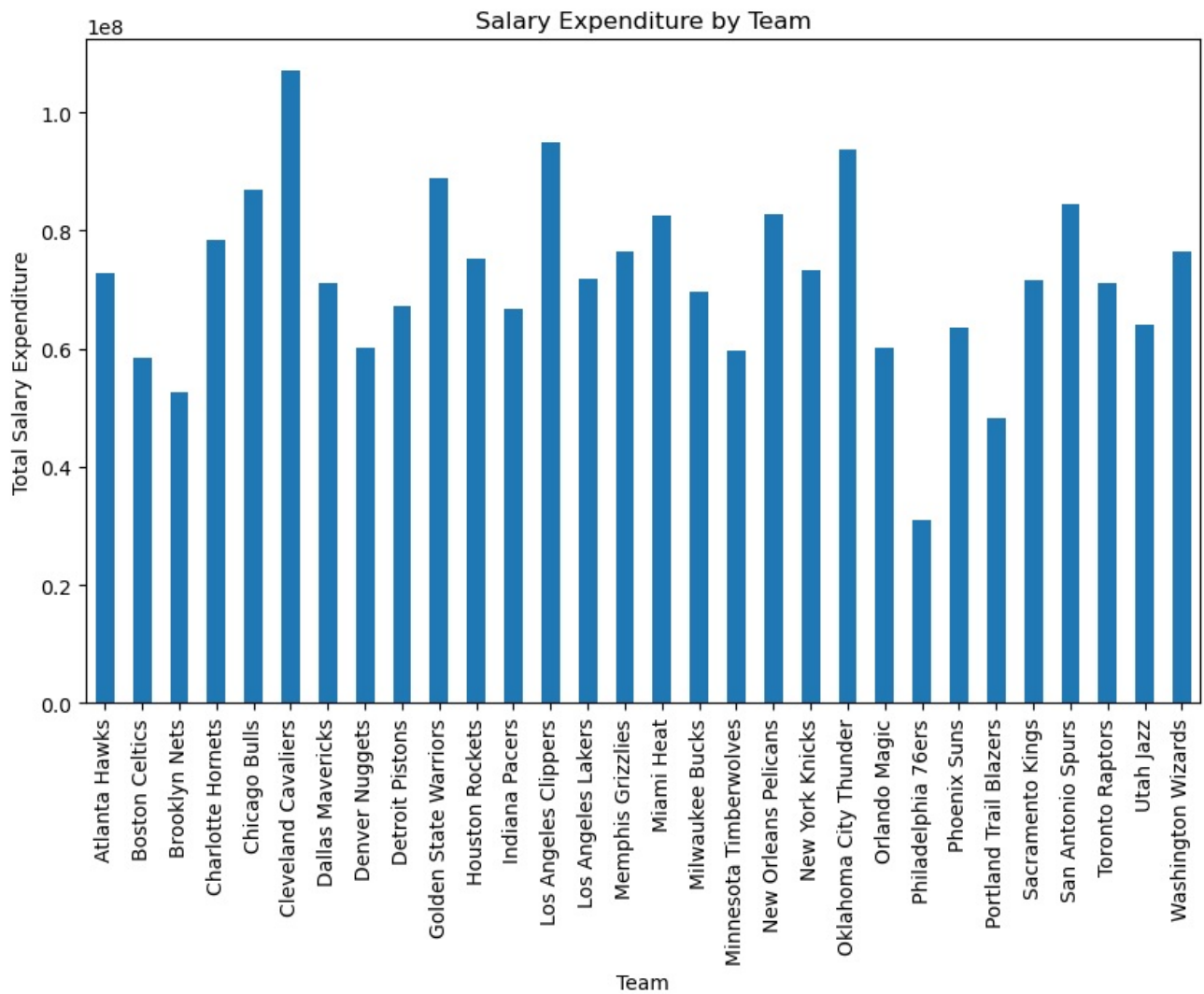
# Find the team and position with the highest salary expenditure
highest_salary_team = salary_by_team.idxmax()
highest_salary_position = salary_by_position.idxmax()

highest_salary_team, highest_salary_position
```

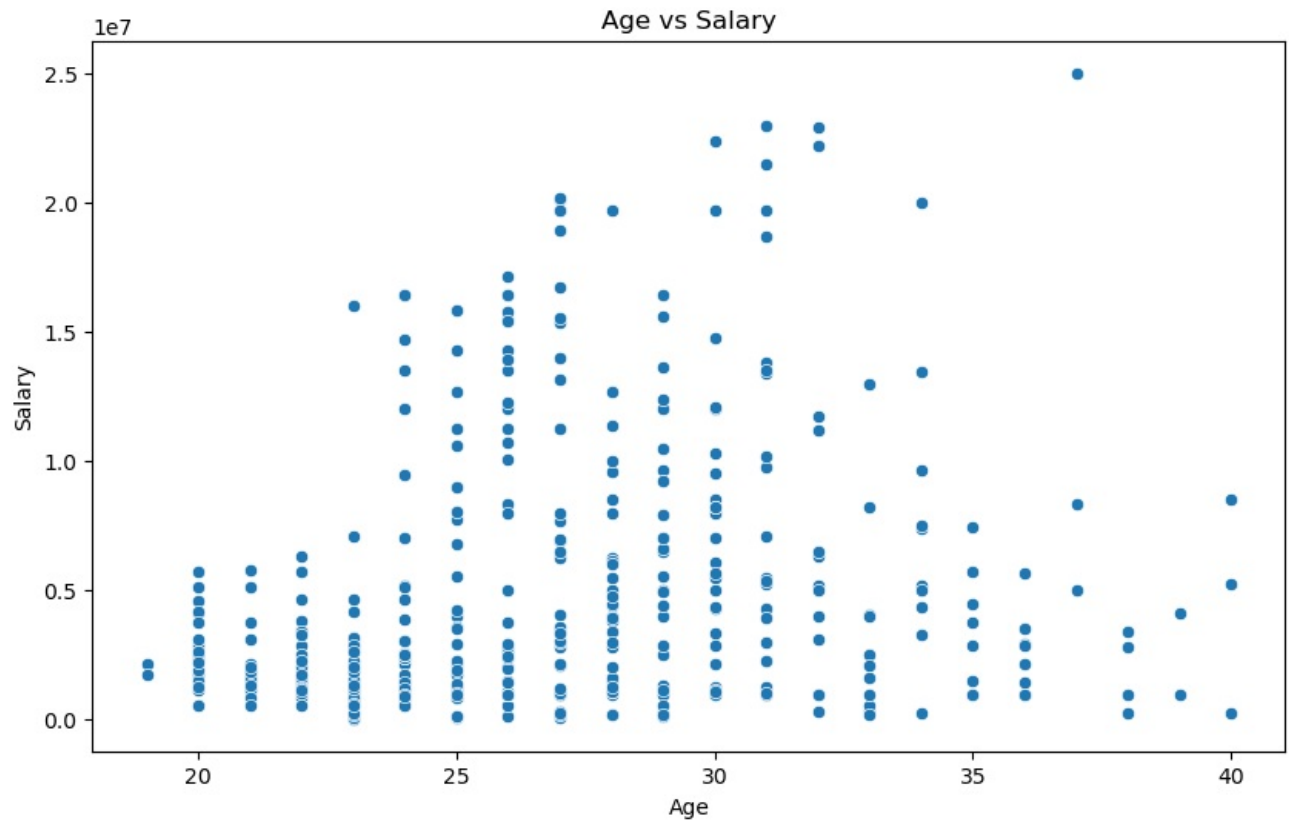
```
Out[10]: ('Cleveland Cavaliers', 'C')
```

```
In [11]: plt.figure(figsize=(10, 6))
salary_by_team.plot(kind='bar')
plt.title('Salary Expenditure by Team')
plt.ylabel('Total Salary Expenditure')
plt.xlabel('Team')
plt.show()

plt.figure(figsize=(10, 6))
salary_by_position.plot(kind='bar')
plt.title('Salary Expenditure by Position')
plt.ylabel('Total Salary Expenditure')
plt.xlabel('Position')
plt.show()
```



```
In [12]: plt.figure(figsize=(10, 6))
sns.scatterplot(x=data['Age'],y=data['Salary'])
plt.title('Age vs Salary')
plt.xlabel('Age')
plt.ylabel('Salary')
plt.show()
```



```
In [13]: correlation = data[['Age', 'Salary']].corr()  
correlation
```

```
Out[13]:
```

	Age	Salary
Age	1.000000	0.214009
Salary	0.214009	1.000000

```
In [ ]:
```

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